

Problem

Find the quadrature (cosine and sine) form of the Fourier series

$$f(t) = 7.5 + \sum_{n=1}^{\infty} \frac{37.5}{n^3 + 1} \cos\left(2nt + \frac{n\pi}{4}\right)$$

Step-by-step solution

Step 1 of 2

The amplitude-phase form of the Fourier series is,

$$f(t) = 10 + \sum_{n=1}^{\infty} \frac{37.5}{n^3 + 1} \cos\left(2nt + \frac{n\pi}{4}\right)$$

Expand the cosine function as per the following formula:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Apply the formula to the function, $f(t)$.

$$\begin{aligned} f(t) &= 10 + \sum_{n=1}^{\infty} \frac{37.5}{n^3 + 1} \left\{ \cos(2nt) \cos\left(\frac{n\pi}{4}\right) - \sin(2nt) \sin\left(\frac{n\pi}{4}\right) \right\} \\ &= 10 + \sum_{n=1}^{\infty} \left\{ \frac{37.5}{n^3 + 1} \cos\left(\frac{n\pi}{4}\right) \cos(2nt) - \frac{37.5}{n^3 + 1} \sin\left(\frac{n\pi}{4}\right) \sin(2nt) \right\} \dots (1) \end{aligned}$$

Consider the general form of Fourier series expansion:

$$f(t) = \underbrace{a_0}_{\text{dc component}} + \underbrace{\sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)}_{\text{ac component}} \dots (2)$$

Step 2 of 2

Compare the function in equation (1) with the general form of Fourier series in equation (2).

$$a_0 = 10$$

$$a_n = \frac{37.5}{n^3 + 1} \cos\left(\frac{n\pi}{4}\right)$$

$$b_n = -\frac{37.5}{n^3 + 1} \sin\left(\frac{n\pi}{4}\right)$$

Here,

a_0 is the dc component of the Fourier series expansion

a_n and b_n are the amplitudes of the sinusoids in the ac component.

Thus, the function is in the form of Fourier series expansion.

For Quadrature form, the components, $\cos(2nt)$ and $\sin(2nt)$ should be orthogonal, that is, the phase between the two components must be 90° .

$$f(t) = 10 + \sum_{n=1}^{\infty} \left\{ \frac{37.5}{n^3 + 1} \cos\left(\frac{n\pi}{4}\right) \cos(2nt) - \frac{37.5}{n^3 + 1} \sin\left(\frac{n\pi}{4}\right) \cos\left(2nt + \frac{\pi}{2}\right) \right\}$$

$$\left[\cos\left(\frac{\pi}{2} + \theta\right) = -\sin(\theta) \right]$$

Therefore, the quadrature form of the Fourier series function, $f(t)$ is

$$\boxed{10 + \sum_{n=1}^{\infty} \left\{ \frac{37.5}{n^3 + 1} \cos\left(\frac{n\pi}{4}\right) \cos(2nt) - \frac{37.5}{n^3 + 1} \sin\left(\frac{n\pi}{4}\right) \cos\left(2nt + \frac{\pi}{2}\right) \right\}}.$$